

INFO251_Gomez_Laura_PS2

February 6, 2024

1 Problem Set 2

1.1 Introduction to the assignment

For this assignment, you will be using data from the [Progresa program](#), a government social assistance program in Mexico. This program, as well as the details of its impact, are described in the paper “[School subsidies for the poor: evaluating the Mexican Progresa poverty program](#)”, by Paul Shultz (available on bCourses). Please familiarize yourself with the PROGRESA program before beginning this problem set, so you have a rough sense of where the data come from and how they were generated. If you just proceed into the problem set without understanding Progresa or the data, it will be very difficult!

The goal of this problem set is to implement some of the basic econometric techniques that you are learning in class to measure the impact of Progresa on secondary school enrollment rates. The timeline of the program was:

- Baseline survey conducted in 1997
- Intervention begins in 1998, “Wave 1” of surveys conducted in 1998
- “Wave 2” of surveys conducted in 1999
- Evaluation ends in 2000, at which point the control villages were treated.

When you are ready, download the `progresa_sample.csv` data from bCourses. The data are actual data collected to evaluate the impact of the Progresa program. In this file, each row corresponds to an observation taken for a given child for a given year. There are two years of data (1997 and 1998), and just under 40,000 children who are surveyed in each year. For each child-year observation, the following variables are collected:

Variable name	Description
year	year in which data is collected
sex	male = 1
indig	indigenous = 1
dist_sec	nearest distance to a secondary school
sc	enrolled in school in year of survey
grc	grade enrolled
fam_n	family size
min_dist	min distance to an urban center
dist_cap	min distance to the capital
poor	poor = 1
progresa	treatment =1

Variable name	Description
hohedu	years of schooling of head of household
hohwag	monthly wages of head of household
welfare_index	welfare index used to classify poor
hohsex	gender of head of household (male=1)
hohage	age of head of household
age	years old
folnum	individual id
village	village id
sc97	schooling in 1997
grc97	grade enrolled in 1997

1.2 Part 1: Descriptive analysis

1.2.1 1.1 Summary Statistics

Present summary statistics (mean, median and standard deviation) for all of the demographic variables in the dataset (i.e., everything except year, folnum, village). Present these in a single table alphabetized by variable name. Do NOT simply expect the grader to scroll through your output!

Note: For this and subsequent problems, you will need to be careful in how you deal with missing (NULL) values. You should not blindly drop rows and columns where any data field is missing. For instance, in calculating the average `hohwag`, you should average the `hohwag` values from all households that report a value (even if the household does not have a recorded `age` value, for example).

```
[ ]: # your code here
import pandas as pd
import numpy as np

# Import progres data set
progres_df = pd.read_csv('progres_sample.csv')
progres_df.head()

#clean up columns that are not numerical
progres_df['poor'] = progres_df['poor'].eq('pobre').astype(int)
progres_df['progres'] = progres_df['progres'].eq('basal').astype(int)

# define and fill summary table
progres_summary = pd.DataFrame(index = ['Mean', 'Median',
    ↪ 'Standard_Deviation'], columns = progres_df.columns)
progres_summary.loc['Mean'] = progres_df.mean(axis = 0, numeric_only = True)
progres_summary.loc['Median'] = progres_df.median(axis = 0, numeric_only =
    ↪ True)
```

```

progresasummary.loc['Standard Deviation'] = progresasummary.std(axis = 0,
↳ numeric_only = True)

# drop variables and sort summary df
vars_to_exclude = ['year', 'folnum', 'village']
progresasummary = progresasummary.drop(columns = vars_to_exclude)
progresasummary = progresasummary.sort_index(axis=1)
progresasummary.T

```

```

[ ]:

```

	Mean	Median	Standard Deviation
age	11.36646	11.0	3.167744
dist_cap	147.674452	132.001494	76.063134
dist_sec	2.41891	2.279	2.234109
fam_n	7.215715	7.0	2.3529
grc	3.963537	4.0	2.499063
grc97	3.705372	4.0	2.572387
hohage	44.436717	43.0	11.620372
hohedu	2.768104	2.0	2.656106
hohsex	0.925185	1.0	0.263095
hohwag	586.985312	500.0	788.133664
indig	0.298324	0.0	0.457525
min_dist	103.44752	111.228612	42.089441
poor	0.846498	1.0	0.360473
progresas	0.615663	1.0	0.486441
sc	0.819818	1.0	0.384342
sc97	0.813922	1.0	0.389172
sex	0.512211	1.0	0.499854
welfare_index	690.346564	685.0	139.49113

1.2.2 1.2 Differences at baseline?

Are the baseline (1997) demographic characteristics **for the poor** different in treatment and control villages? Present your results in a single table with the following columns and 14 (or so) rows (alphabetized by variable name):

Variable name	Average value (Treatment villages)	Average value (Control villages)	Difference (Treat - Control)	p-value
Male	?	?	?	?

Hint: Use a T-Test to determine whether there is a statistically significant difference in the average values of each of the variables in the dataset. Focus only on the data from 1997 from poor households (i.e., poor=='pobre').

```

[ ]: from scipy.stats import ttest_ind

#initialize summary df to fill

```

```

vars_to_exclude = ['progresas', 'poor']
poor_df_summary = pd.DataFrame(columns = ['mean_val_treatment',
    ↳ 'mean_val_control', 'difference', 'p_value'], index = progresas_summary.
    ↳ columns)
poor_df_summary = poor_df_summary.rename_axis('variable_name', axis = 0)
poor_df_summary = poor_df_summary.drop(vars_to_exclude)

# reduce data set to look at just 1997 and poor households
poor_df = progresas_df[(progresas_df['poor'] == 1) & (progresas_df['year'] == 97)]

# fill out poor_df summary
poor_df_summary['mean_val_treatment'] = poor_df[poor_df['progresas'] == 1].
    ↳ mean(skipna=True)
poor_df_summary['mean_val_control'] = poor_df[poor_df['progresas'] == 0].
    ↳ mean(skipna=True)
poor_df_summary['difference'] = poor_df_summary['mean_val_treatment'] -
    ↳ poor_df_summary['mean_val_control']

# perform independent t-test to get p_value
for var in poor_df_summary.index:
    _, poor_df_summary.loc[var, 'p_value'] =
    ↳ ttest_ind(poor_df[poor_df['progresas'] == 1][var].dropna(),
    ↳ poor_df[poor_df['progresas'] == 0][var].dropna())
poor_df_summary

```

```

[ ]:
      mean_val_treatment  mean_val_control  difference  p_value
variable_name
age                    10.716991         10.742023   -0.025032  0.478559
dist_cap               150.829074        153.769730   -2.940656  0.000842
dist_sec                2.453122         2.507662   -0.054540  0.035698
fam_n                   7.281327         7.302469   -0.021142  0.427104
grc                     3.531599         3.543050   -0.011450  0.689015
grc97                   3.531599         3.543050   -0.011450  0.689015
hohage                  43.648828         44.276918   -0.628090  0.000002
hohedu                  2.663139         2.590348    0.072791  0.011051
hohsex                  0.924656         0.922947    0.001709  0.571186
hohwag                  544.339544        573.163558  -28.824015  0.000325
indig                   0.325986         0.332207   -0.006222  0.24536
min_dist               107.152915        103.237854    3.915060    0.0
sc                      0.822697         0.815186    0.007511  0.095178
sc97                   0.822697         0.815186    0.007511  0.095178
sex                    0.519317         0.505052    0.014265  0.012192
welfare_index          655.428377        659.579100   -4.150723  0.001431

```

1.2.3 1.3 Interpretation

- A: Are there statistically significant differences between treatment and control villages at baseline?
- B: Why does it matter if there are differences at baseline?
- C: What does this imply about how to measure the impact of the treatment?

Discuss your results here 1. Yes, there are statistically significant differences between some of the treatment and control villages at baseline. Considering an alpha value of 0.01, so that $p_value < 0.01$ is considered significant, we find that `dist_cap`, `hohage`, `hohwag`, `min_dist` and `welfare_index` are all significantly different at baseline. 2. It matters that there are differences at baseline since under randomized treatment, we hope a simple difference between outcomes in treated and control units will give an unbiased estimate of impact. In other words, we want to attribute any changes from *progesa* treatment to be due to the intervention and not due to some difference that was present at baseline that may interfere with the interpretation of the impact. 3. What this implies is that when we are assessing the measure of the impact of *progesa*, we need to control for these variables (regress out their contribution) in the pre/post treatment analysis to get an unbiased estimate of the impact of treatment.

1.2.4 1.4 Graphical exploration, part 1

Visualize the distribution of village enrollment rates **among poor households in treated villages**, before and after treatment. Specifically, for each village, calculate the average rate of enrollment of poor households in treated villages in 1997, then compute the average rate of enrollment of poor households in treated villages in 1998. Create two separate histograms showing the distribution of these average enrollments rates, one histogram for 1997 and one histogram for 1998. On each histogram, draw a vertical line that intersects the x-axis at the average value (across all households). Does there appear to be a difference? Is this difference statistically significant?

```
[ ]: # Generate 2 distributions
import seaborn as sns
import matplotlib.pyplot as plt
sns.set_theme(style="ticks", context='notebook')
cmap = sns.color_palette()

# look at poor, enrolled villages only
poor_treated_villages = progresas_df[(progresas_df['poor'] == 1) &
    ↪ (progresas_df['progresas'] == 1)]
village_ids_treated = np.unique(poor_treated_villages['village']) # treated
    ↪ village ids

# define pre and post treatment distributions
pre_treatment_dist = pd.DataFrame(index = village_ids_treated, columns =
    ↪ ['mean_enrollment'])
post_treatment_dist = pd.DataFrame(index = village_ids_treated, columns =
    ↪ ['mean_enrollment'])

# iterate through villages to get mean enrollment
```

```

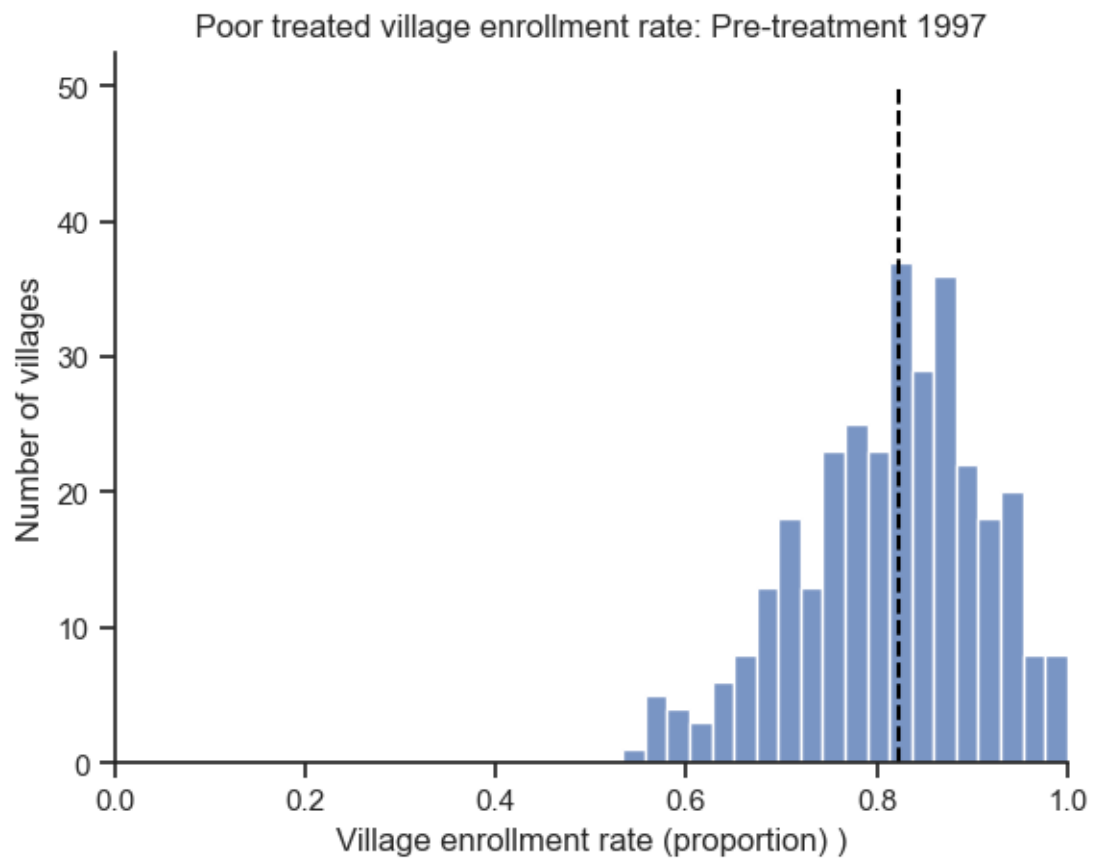
for village in village_ids_treated:
    pre_treatment_dist.loc[village, 'mean_enrollment'] =
    ↪poor_treated_villages[(poor_treated_villages['village'] == village) &
    ↪(poor_treated_villages['year'] == 97)][ 'sc'].mean()
    post_treatment_dist.loc[village, 'mean_enrollment'] =
    ↪poor_treated_villages[(poor_treated_villages['village'] == village) &
    ↪(poor_treated_villages['year'] == 98)][ 'sc'].mean()

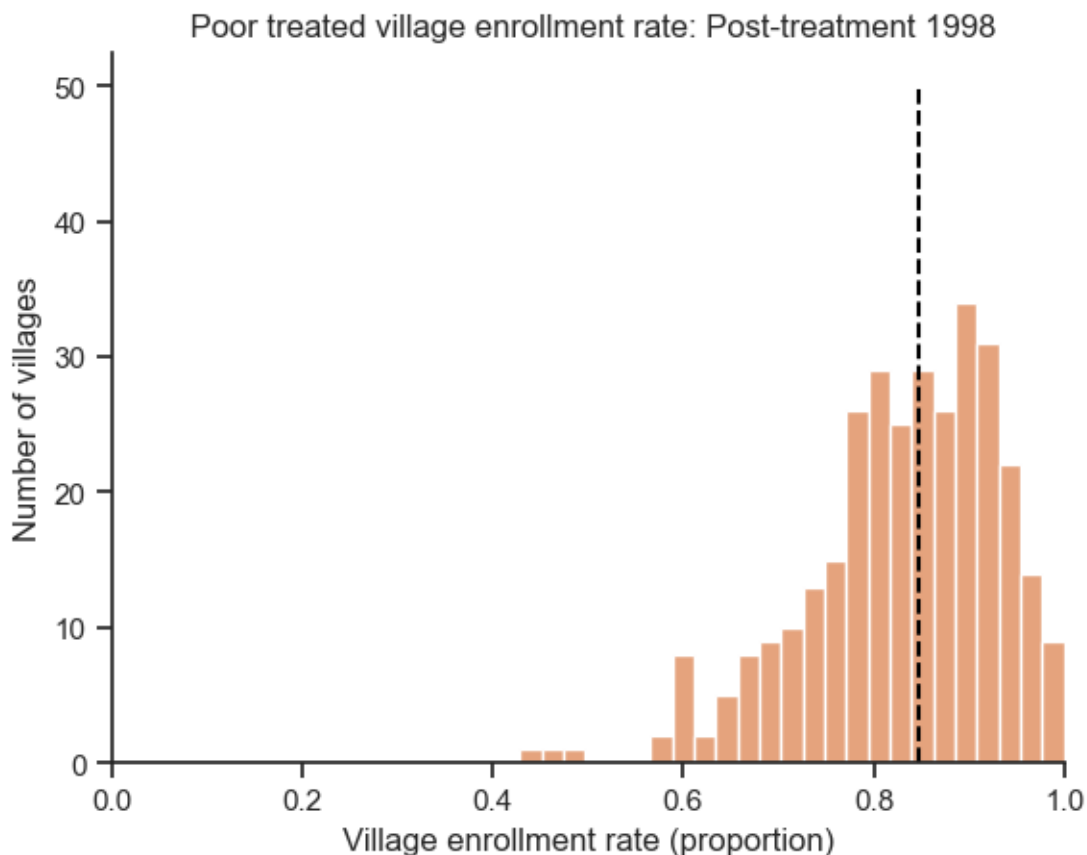
# make sure its numeric
pre_treatment_dist = pre_treatment_dist.astype('float64')
post_treatment_dist = post_treatment_dist.astype('float64')

# Plot histograms
xlims = [0,1]
sns.histplot(pre_treatment_dist['mean_enrollment'], bins = 20, color = cmap[0])
mean_pre_treatment = poor_treated_villages[poor_treated_villages['year'] ==
    ↪97][ 'sc'].mean() # take mean across households, not across villages
plt.vlines(mean_pre_treatment, 0, 50, color = 'black', linestyle='dashed')
plt.xlim(xlims)
plt.ylabel('Number of villages')
plt.xlabel('Village enrollment rate (proportion) ')
plt.title(' Poor treated village enrollment rate: Pre-treatment 1997 ')
sns.despine()
plt.show()

sns.histplot(post_treatment_dist['mean_enrollment'], bins = 25, color = cmap[1])
mean_post_treatment = poor_treated_villages[poor_treated_villages['year'] ==
    ↪98][ 'sc'].mean() # take mean across households, not across villages
plt.vlines(mean_post_treatment, 0, 50, color = 'black', linestyle='dashed')
plt.xlim(xlims)
plt.ylabel('Number of villages')
plt.xlabel('Village enrollment rate (proportion)')
plt.title(' Poor treated village enrollment rate: Post-treatment 1998')
sns.despine()
plt.show()

```





```
[ ]: ## Part 2: check if difference is significant

# Compute statistic on individual households
pre_treatment_sc = poor_treated_villages[poor_treated_villages['year'] == 97]['sc'].dropna()
post_treatment_sc = poor_treated_villages[poor_treated_villages['year'] == 98]['sc'].dropna()
_, pval_enrollment_treatment_sc = ttest_ind(pre_treatment_sc, post_treatment_sc)
print('Enrollment difference in treatment group (households): ' + str(pval_enrollment_treatment_sc))

# Compute statistic on villages
_, pval_enrollment_treatment = ttest_ind(post_treatment_dist['mean_enrollment'], pre_treatment_dist['mean_enrollment'])
print('Enrollment difference in treatment group (villages): ' + str(pval_enrollment_treatment))
```

```
Enrollment difference in treatment group (households): 1.2976490728161e-09
Enrollment difference in treatment group (villages): 0.04424870036572114
```

Does there appear to be a difference? Is this difference statistically significant? - There does appear to be difference in the distribution of mean enrollment across villages in the post vs pre treatment years which can be visually seen by comparing where the dashed line intersects the x-axis in each histogram. - When computing a t-test across all households for difference in enrollment in the pre/post period, we obtain a pvalue $\ll 0.01$, which suggests that there is a significant difference in enrollment rates after treatment in the poor villages.

1.2.5 1.5 Graphical exploration, part 2

Repeat the above exercise for poor households in **control villages**, before and after treatment. Do you observe a difference in enrollment in control villages between 1997 and 1998? How does what you observe here affect how you might approach measuring the impact of PROGRESA?

```
[ ]: # look at poor, not enrolled villages only
poor_control_villages = progresas_df[(progresas_df['poor'] == 1) &
    ↪(progresas_df['progresas'] == 0)]
village_ids_ctrl = np.unique(poor_control_villages['village']) # treated
    ↪villages

# define pre and post treatment distributions
pre_treatment_ctrl = pd.DataFrame(index = village_ids_ctrl, columns =
    ↪['mean_enrollment'])
post_treatment_ctrl = pd.DataFrame(index = village_ids_ctrl, columns =
    ↪['mean_enrollment'])

# iterate through villages to get mean enrollment
for village in village_ids_ctrl:
    pre_treatment_ctrl.loc[village, 'mean_enrollment'] =
    ↪poor_control_villages[(poor_control_villages['village'] == village) &
    ↪(poor_control_villages['year'] == 97)][['sc']].mean()
    post_treatment_ctrl.loc[village, 'mean_enrollment'] =
    ↪poor_control_villages[(poor_control_villages['village'] == village) &
    ↪(poor_control_villages['year'] == 98)][['sc']].mean()

# make sure its numeric
pre_treatment_ctrl = pre_treatment_ctrl.astype('float64')
post_treatment_ctrl = post_treatment_ctrl.astype('float64')

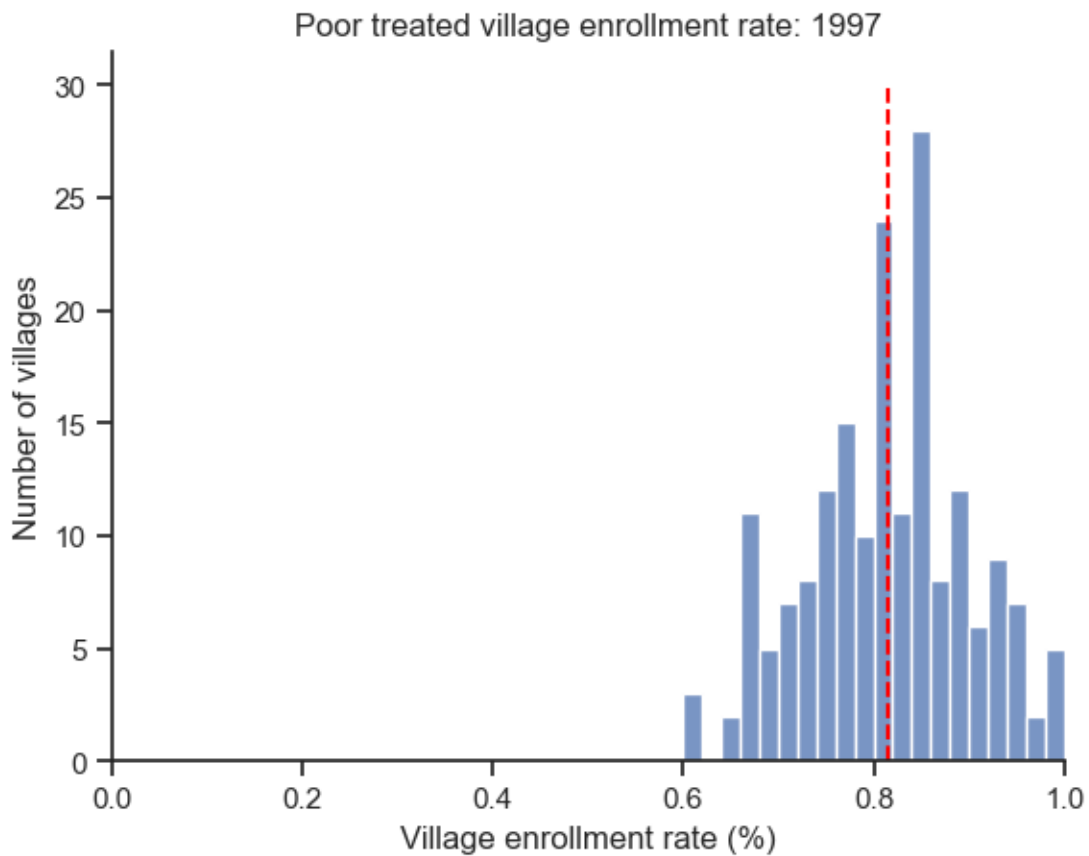
# Plot histograms
sns.histplot(pre_treatment_ctrl['mean_enrollment'], bins = 20, color = cmap[0])
mean_pre_interv = poor_control_villages[poor_control_villages['year'] ==
    ↪97][['sc']].mean() # take mean across households, not across villages
plt.vlines(mean_pre_interv, 0, 30, color = 'red', linestyle='dashed')
plt.xlim(xlims)
plt.ylabel('Number of villages')
plt.xlabel('Village enrollment rate (%)')
plt.title('Poor treated village enrollment rate: 1997')
```

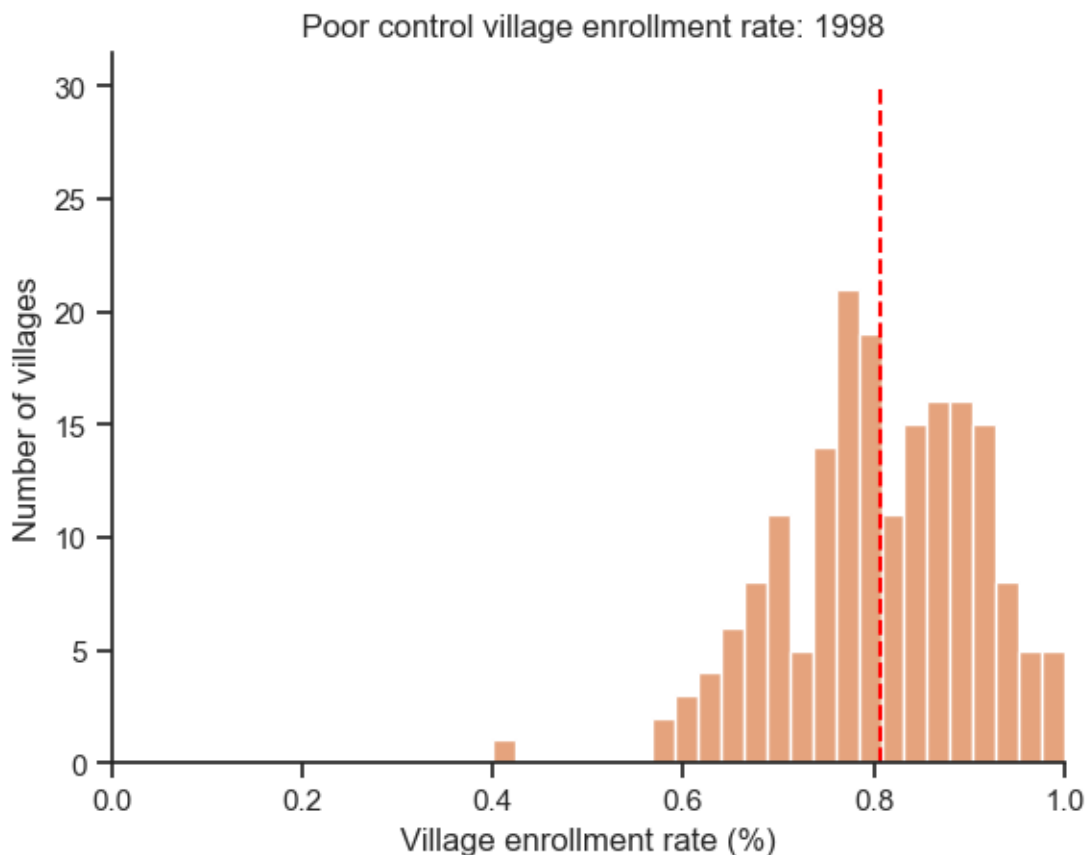
```

sns.despine()
plt.show()

sns.histplot(post_treatment_ctrl['mean_enrollment'], bins = 25, color = cmap[1])
mean_post_interv = poor_control_villages[poor_control_villages['year'] == 1998][
    'sc'].mean() # take mean across households, not across villages
plt.vlines(mean_post_interv, 0, 30, color = 'red', linestyle='dashed')
plt.xlim(xlims)
plt.ylabel('Number of villages')
plt.xlabel('Village enrollment rate (%)')
plt.title(' Poor control village enrollment rate: 1998')
sns.despine()
plt.show()

```





```
[ ]: ## Part 2: check if difference is significant

# Compute statistic on individual households
pre_treatment_ctrl_sc = poor_control_villages[poor_control_villages['year'] == 1997]['sc'].dropna()
post_treatment_ctrl_sc = poor_control_villages[poor_control_villages['year'] == 1998]['sc'].dropna()
_, pval_enrollment_control_sc = ttest_ind(pre_treatment_ctrl_sc, post_treatment_ctrl_sc)
print('Enrollment difference in control group (households): ' + str(pval_enrollment_control_sc))

# Compute statistic on villages
_, pval_enrollment_control = ttest_ind(pre_treatment_ctrl['mean_enrollment'], post_treatment_ctrl['mean_enrollment'])
print('Enrollment difference in control group (villages): ' + str(pval_enrollment_control))
```

Enrollment difference in control group (households): 0.1511345297015608

Enrollment difference in control group (villages): 0.4349129201749543

Do you observe a difference in enrollment in control villages between 1997 and 1998? How does what you observe here affect how you might approach measuring the impact of PROGRESA? - By performing a t-test comparing enrollment for control villages not receiving progressa in 1997 vs 1998, we find that the $p_value = 0.151$ across households, which does not meet the criterion value of $p = 0.01$. Therefore, there is no significant difference in enrollment in the control group between these time points. - What this reveals is that we may have done proper randomization when assigning control vs treatment since we would expect not to find a difference in enrollment across time if there was no intervention. Therefore, when we explore the effect of progressa on enrollment we may be able to use a simple pre vs post estimator there may not be a confound of a time trend.

1.3 Part 2: Measuring Impact

Our goal is to estimate the causal impact of the PROGRESA program on the social and economic outcomes of individuals in Mexico. We will focus on the impact of the program on school enrollment rates among the poor (those with `poor == 'pobre'`), since only the poor were eligible to receive PROGRESA assistance, and since a primary objective of the program was to increase school enrollment.

1.3.1 2.1 Simple differences: T-test

Begin by estimating the impact of Progresa using “simple differences.” Restricting yourself to data from 1998 (after treatment), calculate the average enrollment rate among **poor** households in the Treatment villages and the average enrollment rate among **poor** households in the control villages. Use a t-test to determine if this difference is statistically significant. What do you conclude?

```
[ ]: # Your code here

# get df of enrollment restricted to poor villages with or without treatment
# in the year 1998
post_ctrl_df = progres_a_df[(progres_a_df['poor'] == 1) &
    ↳ (progres_a_df['progres_a'] == 0) & (progres_a_df['year'] == 98)]
post_treatment_df = progres_a_df[(progres_a_df['poor'] == 1) &
    ↳ (progres_a_df['progres_a'] == 1) & (progres_a_df['year'] == 98)]

# calculate mean enrollment among households
post_ctrl_mean_enrollment = post_ctrl_df['sc'].mean()
post_trt_mean_enrollment = post_treatment_df['sc'].mean()

# calculate difference: Treatment - Control
diff = post_trt_mean_enrollment - post_ctrl_mean_enrollment
print('Simple differences: Treatment - Control in 1998: ' + str(diff))

# perform t-test to determine significance
_, pval_enrollment_diff = ttest_ind(post_ctrl_df['sc'].dropna(),
    ↳ post_treatment_df['sc'].dropna())
```

```
print('Simple differences p value : Treatment - Control in 1998: ' +
      str(pval_enrollment_diff))
```

Simple differences: Treatment - Control in 1998: 0.0388421646651228

Simple differences p value : Treatment - Control in 1998: 6.636344447523235e-17

Discuss your results here - In this simple treatment vs control estimator we find there is a significant difference between treatment and control groups in 1998 (after progresra intervention). Because of the small magnitude of the p-value, this strongly suggests there is a correlation between treatment intervention and enrollment rate of children. If given proper randomization, the Beta term should be zero and this would be an unbiased estimate.

1.3.2 2.2 Simple differences: Regression

Estimate the effects of Progresra on enrollment using a regression model, by regressing the 1998 enrollment rates **of the poor** on treatment assignment. For now, do not include any other variables in your regression. Discuss the following:

- Based on this model, how much did Progresra increase or decrease the likelihood of a child enrolling? Make sure you express your answer in a sentence that a person with no technical background could understand, using appropriate units.
- How does your regression estimate compare to your t-test estimate from part 2.1?
- Based on this regression model, can we reject the null hypothesis that the treatment effects are zero?
- What is the counterfactual assumption underlying this regression?

```
[ ]: # Your code here
import statsmodels.api as sm

# constrict analysis on poor population and for the year 1998
regression_df = progresra_df[(progresra_df['poor'] == 1) & (progresra_df['year'] == 98)]
regression_df_simple = regression_df[['progresra', 'sc']].dropna(axis=0) # only drop nans for individuals with missing data for enrollment or progresra

# assign independent and dependent variables
X = sm.add_constant(regression_df_simple['progresra'])
y = regression_df_simple['sc']
ols = sm.OLS(y, X)

# run the regression
ols_result = ols.fit()
ols_result.summary()
```

[]:

Dep. Variable:	sc	R-squared:	0.003
Model:	OLS	Adj. R-squared:	0.003
Method:	Least Squares	F-statistic:	69.87
Date:	Tue, 06 Feb 2024	Prob (F-statistic):	6.64e-17
Time:	01:45:14	Log-Likelihood:	-11926.
No. Observations:	27450	AIC:	2.386e+04
Df Residuals:	27448	BIC:	2.387e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	0.8076	0.004	220.676	0.000	0.800	0.815
progres	0.0388	0.005	8.359	0.000	0.030	0.048

Omnibus:	7638.939	Durbin-Watson:	1.734
Prob(Omnibus):	0.000	Jarque-Bera (JB):	15767.534
Skew:	-1.767	Prob(JB):	0.00
Kurtosis:	4.140	Cond. No.	3.01

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
[ ]: print('T-test pvalue:' + str(ols_result.pvalues[1]))
```

T-test pvalue:6.636344447488953e-17

Discuss your results here 1. Based on this model, how much did Progres increase or decrease the likelihood of a child enrolling? Make sure you express your answer in a sentence that a person with no technical background could understand, using appropriate units. - Children that received progres had a 3.9% higher likelihood of enrolling in school. 2. How does your regression estimate compare to your t-test estimate from part 2.1? - By taking a look at the p value based on the t-statistic in the statsmodel summary, we find that the p-value is the same as that computed using a t-test from part 2.1 -> p_val = 6.63E-17 3. Based on this regression model, can we reject the null hypothesis that the treatment effects are zero? - Yes, based on this model since the p_value is < 0.01, we can reject that the treatment effects are zero and that there is a strong correlation between receiving progressa and rate of enrollment. 4. What is the counterfactual assumption underlying this regression? - The counterfactual assumption here is that the control group (poor villages that didn't receive treatment) enrollment rates in 1998 represent the enrollment rates in 1998 that would have occurred had the treated group (poor villages that received progressa), not actually received any intervention. Specifically, the counterfactual assumption is that in the absence of receiving progres, the villages that received treatment would have experienced similar enrollment rates to those that did not receive progres.

1.3.3 2.3 Multiple Regression

Estimate the above regression, but this time include a set of control variables. Include, for instance, age, distance to a secondary school, gender, education of household head, welfare index, indigenous, etc.

- How do the controls affect the point estimate of treatment effect?
- How do the controls affect the standard error on the treatment effect?

- How do you interpret the differences (or similarities) between your estimates of 2.2 and 2.3?
- Interpret the coefficient associated with the `dist_sec` variable. Is this evidence that the household's distance from a secondary school has a *causal* impact on educational attainment?

```
[ ]: # Your code here
# define variables of interest from regression_df ( poor population in year
↳1998)
regression_df_multiple = regression_df[['progresa', 'dist_sec', 'sex', 'age',
↳'sc']].dropna(axis=0) # only drop nans for individuals with missing data for
↳enrollment or progresa

# assign independent and dependent variables
X = sm.add_constant(regression_df_multiple[['progresa', 'dist_sec', 'sex',
↳'age']])
y = regression_df_multiple['sc'].values
ols = sm.OLS(y, X)

# run the regression
ols_multiple_result = ols.fit()
ols_multiple_result.summary()
```

[]:

Dep. Variable:	y	R-squared:	0.261
Model:	OLS	Adj. R-squared:	0.261
Method:	Least Squares	F-statistic:	2423.
Date:	Mon, 05 Feb 2024	Prob (F-statistic):	0.00
Time:	23:23:10	Log-Likelihood:	-7805.5
No. Observations:	27440	AIC:	1.562e+04
Df Residuals:	27435	BIC:	1.566e+04
Df Model:	4		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.5706	0.009	177.798	0.000	1.553	1.588
progres	0.0357	0.004	8.927	0.000	0.028	0.044
dist_sec	-0.0123	0.001	-14.229	0.000	-0.014	-0.011
sex	0.0307	0.004	7.905	0.000	0.023	0.038
age	-0.0662	0.001	-96.944	0.000	-0.068	-0.065

Omnibus:	3068.975	Durbin-Watson:	1.703
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4208.713
Skew:	-0.950	Prob(JB):	0.00
Kurtosis:	3.267	Cond. No.	55.3

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Discuss your results here 1. How do the controls affect the point estimate of treatment effect? - Considering the control variables age, sex and distance to secondary school; they seem to have little effect on the point estimate of the treatment effect, that being the coefficient for the *progresa* regressor in this multiple regression (0.0357 MR vs 0.0388 in the SR). 2. How do the controls affect

the standard error on the treatment effect? - The controls act to reduce the absolute value of the standard error on the treatment effect. Taking into account the reduction in the point estimate of the treatment effect (ie: comparing the t-statistic), we find that control variables result in a higher t-statistic for *progesa* and thereby a higher significance of the *progesa* point estimate. 3. How do you interpret the differences (or similarities) between your estimates of 2.2 and 2.3? - There is a high-level of similarity between the *progesa* estimate from the simple regression results in 2.2 and multiple regression results in 2.3, suggesting that the treatment effect is robust to the inclusion of additional controls and co-variables. In other words, the treatment effect is unlikely to be biased by confounding factors such as the control variables included. 4. Interpret the coefficient associated with the `dist_sec` variable. Is this evidence that the household's distance from a secondary school has a *causal* impact on educational attainment? - The coefficient for `dist_sec` was significant with small standard errors. The t-statistic had a large magnitude in comparison with the *progesa* variable, thereby suggesting that there is a large correlation between the enrollment rate of a household and the distance from the nearest secondary school. This makes logical sense when considering the context of the variable as we would expect it to have high predictability on enrollment. Further rigorous estimation such as a difference-in-difference regression is needed to test if this impact is causal. For now the correlation is strong.

1.3.4 2.4 Multiple Regression Revisited

For the same set of control variables that you used in 2.3, carry out the following alternative estimation procedure.

- First, regress the 1998 enrollment of the poor on the control variables, **without including the treatment assignment**.
- Second, use this model to obtain predicted values of the 1998 enrollment for each child in the sample used to estimate the model in step 1.
- Third, compute a new value for each child, which is the difference between the actual 1998 enrollment and the predicted enrollment value from step 2.
- Finally, regress the difference from step 3 on treatment assignment.

Compare the point estimate and the standard error on treatment assignment that you obtained in step 4 to their analogues in 2.3. Explain the results.

```
[ ]: # Your code here

# 1. regress 1998 enrollement on control varliables without treatment_
↪assignment
regression_df_noprogressa = regression_df[['dist_sec', 'sex', 'age', 'sc']].
↪dropna(axis=0) # only drop nans for individuals with missing data for_
↪enrollment or progesa
X = sm.add_constant(regression_df_noprogressa[['dist_sec', 'sex', 'age']])
y = regression_df_noprogressa['sc'].values
ols = sm.OLS(y, X)
ols_noprogressa_result = ols.fit()
ols_noprogressa_result.summary()

# 2. obtain predicted enrollment using above model
```

```

predicted_enrollment = ols_noprogressa_result.params['const'] +_
↳ols_noprogressa_result.
↳params['dist_sec']*regression_df_noprogressa['dist_sec'] +_
↳ols_noprogressa_result.params['sex']*regression_df_noprogressa['sex'] +_
↳ols_noprogressa_result.params['age']*regression_df_noprogressa['age']

# 3. Compute new value: actual enrolment - predicted enrolment
diff = regression_df_noprogressa['sc'] - predicted_enrollment

# 4. regress difference on treatment assignment

X = sm.add_constant(regression_df_multiple['progresa'])
y = diff.values
ols = sm.OLS(y, X)
ols_diff_result = ols.fit()
ols_diff_result.summary()

```

[]:

Dep. Variable:	y	R-squared:	0.003
Model:	OLS	Adj. R-squared:	0.003
Method:	Least Squares	F-statistic:	79.69
Date:	Tue, 06 Feb 2024	Prob (F-statistic):	4.65e-19
Time:	00:22:32	Log-Likelihood:	-7805.5
No. Observations:	27440	AIC:	1.561e+04
Df Residuals:	27438	BIC:	1.563e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-0.0222	0.003	-7.031	0.000	-0.028	-0.016
progresa	0.0357	0.004	8.927	0.000	0.028	0.044
Omnibus:	3063.571	Durbin-Watson:	1.703			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	4199.137			
Skew:	-0.949	Prob(JB):	0.00			
Kurtosis:	3.266	Cond. No.	3.01			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Discuss your results here - The point estimate and the standard errors on treatment assignment are the exact same as in 2.3. This makes sense since what we are essentially doing with this estimation procedure is seeing how well the left over difference observed between the actual and predicted enrollment rates from the reduced multiple regression is explained by the treatment assignment when we regress the difference with treatment assignment. Its a good way of checking our understanding of how each regressor is contributing in a multiple regression analysis.

1.3.5 2.5 Difference-in-Difference, version 1 (tabular)

Thus far, we have computed the effects of Progresa by estimating the difference in 1998 enrollment rates across villages. An alternative approach would be to compute the treatment effect using a

difference-in-differences framework.

Begin by estimating the average treatment effects of the program for poor households using data from 1997 and 1998. Specifically, calculate the difference (between 1997 and 1998) in enrollment rates among poor households in treated villages; then compute the difference (between 1997 and 1998) in enrollment rates among poor households in control villages.

Display your results in a 2x2 table where the rows are Control/Treatment and the columns are 1997/1998.

- What is your difference-in-difference estimate of the impact, and how does it compare to your earlier (simple difference) results?
- What is the counterfactual assumption underlying this estimate?

```
[ ]: # Your code here
# build 2x2 mean enrollment rate table
diffndiff_table = pd.DataFrame(columns = ['1997', '1998'], index = ['Control',
    ↪ 'Treatment'])
diffndiff_table.loc['Control', '1997'] = progres_a_df[(progres_a_df['poor'] == 1)
    ↪ & (progres_a_df['progres_a'] == 0) & (progres_a_df['year'] == 97)]['sc'].mean()
diffndiff_table.loc['Control', '1998'] = progres_a_df[(progres_a_df['poor'] == 1)
    ↪ & (progres_a_df['progres_a'] == 0) & (progres_a_df['year'] == 98)]['sc'].mean()
diffndiff_table.loc['Treatment', '1997'] = progres_a_df[(progres_a_df['poor'] ==
    ↪ 1) & (progres_a_df['progres_a'] == 1) & (progres_a_df['year'] == 97)]['sc'].
    ↪ mean()
diffndiff_table.loc['Treatment', '1998'] = progres_a_df[(progres_a_df['poor'] ==
    ↪ 1) & (progres_a_df['progres_a'] == 1) & (progres_a_df['year'] == 98)]['sc'].
    ↪ mean()

# calculate difference and estimator
diff_treated = diffndiff_table.loc['Treatment', '1998'] - diffndiff_table.
    ↪ loc['Treatment', '1997']
diff_ctrl = diffndiff_table.loc['Control', '1998'] - diffndiff_table.
    ↪ loc['Control', '1997']

diffndiff_estimate = diff_treated - diff_ctrl

# output results
print('Difference-in-difference estimate of the impact: '+
    ↪ str(diffndiff_estimate))
diffndiff_table.head()
```

Difference-in-difference estimate of the impact: 0.031331280319323085

```
[ ]:          1997      1998
Control    0.815186  0.807637
Treatment  0.822697  0.846479
```

*Discuss your results here 1. What is your difference-in-difference estimate of the impact, and how does it compare to your earlier (simple difference) results? - The difference-in-difference estimate

of the impact is 0.0313. The earlier simple difference computed in 2.1 was 0.038842. The diff-in-diff estimates is smaller in magnitude since it takes into account any time trend common to control and treatment groups and any permanent differences that may exist between the treatment and control groups.

2. What is the counterfactual assumption underlying this estimate?

- The counterfactual assumption here is that in the absence of the treatment, the *progresa* group and the control group would have followed enrollment rates over time. Moreover the mean outcome difference between the treated and control groups before the treatment represents how they would have continued to differ in the absence of the treatment. This would imply that any change in the outcome after the treatment would be due to the treatment assuming that the former trends assumption holds.

1.3.6 2.6 Difference-in-Difference, version 2 (regression)

Now use a regression specification to estimate the average treatment effects of the program in a difference-in-differences, for the poor households. Do this (i) first without including any control variables; and then (ii) do it a second time including at least 5 control variables.

- What is your estimate (i) of the impact of Progresa? Be very specific in interpreting your coefficients and standard errors, and make sure to specify exactly what units you are measuring and estimating.
- Does your estimate of the impact of Progresa from (i) change when you add control variables as in (ii)? How do you explain these changes, or the lack of changes on the *progresa* coefficient between (i) and (ii)?
- How do the estimates from (i) and (ii) compare to the difference-in-difference estimates from 2.4 above? What accounts for these differences, if any exist?
- What is the counterfactual assumption underlying regression (ii)?

```
[ ]: # Your code here
# a) diff in diff for no control variables
regression_df = progresas_df[progresas_df['poor'] == 1]
regression_df_simple = regression_df[['progresas', 'sc', 'year']].dropna(axis=0)
    ↪ # only drop nans for individuals with missing data

# assign independent and dependent variables
regression_df_simple['did'] = regression_df_simple['progresas'] *
    ↪ regression_df_simple['year'] # make new interaction term
X = sm.add_constant(regression_df_simple[['progresas', 'year', 'did']]) # keep
    ↪ year as numeric eg) 97 vs 98 instead of binary 0 = pre-treatment, 1 =
    ↪ post-treatment
y = regression_df_simple['sc']
ols = sm.OLS(y, X)

# run the regression
ols_result = ols.fit()
ols_result.summary()
```

[]:

Dep. Variable:	sc	R-squared:	0.001
Model:	OLS	Adj. R-squared:	0.001
Method:	Least Squares	F-statistic:	28.31
Date:	Tue, 06 Feb 2024	Prob (F-statistic):	2.76e-18
Time:	01:01:39	Log-Likelihood:	-26242.
No. Observations:	58372	AIC:	5.249e+04
Df Residuals:	58368	BIC:	5.253e+04
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.5474	0.497	3.112	0.002	0.573	2.522
progres	-3.0316	0.632	-4.800	0.000	-4.270	-1.794
year	-0.0075	0.005	-1.480	0.139	-0.018	0.002
did	0.0313	0.006	4.835	0.000	0.019	0.044

Omnibus:	15346.988	Durbin-Watson:	1.397
Prob(Omnibus):	0.000	Jarque-Bera (JB):	30608.651
Skew:	-1.711	Prob(JB):	0.00
Kurtosis:	3.937	Cond. No.	5.72e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 5.72e+04. This might indicate that there are strong multicollinearity or other numerical problems.

```
[ ]: # b) diff in diff for no control variables
regression_df_multiple = regression_df[['progres', 'sc', 'year', 'age', 'sex',
↳ 'dist_sec', 'welfare_index', 'hohedu']].dropna(axis=0) # only drop nans for
↳ individuals with missing data

# assign independent and dependent variables
regression_df_multiple['did'] = regression_df_multiple['progres'] *
↳ regression_df_multiple['year'] # make new interaction term
X = sm.add_constant(regression_df_multiple[['progres', 'year', 'did', 'age',
↳ 'sex', 'dist_sec', 'welfare_index', 'hohedu']]) # keep year as numeric eg)
↳ 97 vs 98 instead of binary 0 = pre-treatment, 1 = post-treatment
y = regression_df_multiple['sc']
ols = sm.OLS(y, X)

# run the regression
ols_result_multiple = ols.fit()
ols_result_multiple.summary()
```

[]:

Dep. Variable:	sc	R-squared:	0.275
Model:	OLS	Adj. R-squared:	0.275
Method:	Least Squares	F-statistic:	2754.
Date:	Tue, 06 Feb 2024	Prob (F-statistic):	0.00
Time:	01:01:39	Log-Likelihood:	-16832.
No. Observations:	58200	AIC:	3.368e+04
Df Residuals:	58191	BIC:	3.376e+04
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-1.2160	0.426	-2.857	0.004	-2.050	-0.382
progresa	-2.9979	0.539	-5.559	0.000	-4.055	-1.941
year	0.0279	0.004	6.391	0.000	0.019	0.036
did	0.0309	0.006	5.594	0.000	0.020	0.042
age	-0.0654	0.000	-143.157	0.000	-0.066	-0.064
sex	0.0338	0.003	12.598	0.000	0.029	0.039
dist_sec	-0.0102	0.001	-16.502	0.000	-0.011	-0.009
welfare_index	2.382e-05	1.19e-05	2.002	0.045	5.01e-07	4.71e-05
hohedu	0.0076	0.001	14.163	0.000	0.007	0.009

Omnibus:	5608.624	Durbin-Watson:	1.476
Prob(Omnibus):	0.000	Jarque-Bera (JB):	7408.785
Skew:	-0.872	Prob(JB):	0.00
Kurtosis:	3.120	Cond. No.	3.29e+05

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
[2] The condition number is large, 3.29e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Discuss your results here 1. What is your estimate (i) of the impact of Progres? Be very specific in interpreting your coefficients and standard errors, and make sure to specify exactly what units you are measuring and estimating. - Based on the estimate in (i), we can say there is a strong impact of progres on enrollment rates. In this diff-in-diff regression estimate the “did” coefficient estimates the true effect of the treatment which is 3.3% increased likelihood of enrollment with progres treatment. The standard error is proportionally small (0.6%) and the impact is reflected by the magnitude of the t-statistic which is large in magnitude compared to the other regressors. 2. Does your estimate of the impact of Progres from (i) change when you add control variables as in (ii)? How do you explain these changes, or the lack of changes on the **progres** coefficient between (i) and (ii)? - The impact estimate does change slightly when adding control variables. The progres coefficient decreased slightly from i to ii which indicates that the treatment group specific effect that accounts for average permanent differences between treatment and control group is not as strong when considering multiple control variables which is a good indication that we are properly considering sources of biases that will improve the estimation of the treatment. 3. How do the estimates from (i) and (ii) compare to the difference-in-difference estimates from 2.4 above? What accounts for these differences, if any exist? - The estimate in (i) is the exact same as the difference-in-difference estimates from 2.4. This is expected given its using the same information (pre/post intervention info for both the control and treatment groups). The estimate in (ii) is

slightly smaller compared to the estimate in 2.4 which is a result of including multiple co-variables (control variables) in the regression. By including control variables in the regression, we address potential sources of bias and improve the precision of the estimated treatment effect. The fact there is very little difference is good since it indicates there is a strong effect due to our intervention and not due to some source of variation in the the control variables. 4. What is the counterfactual assumption underlying regression (ii)? - The counterfactual assumption in the second regression is that in the absence of the treatment, the treatment group and the control group would have had similar enrollment rates in 1998. With the inclusion of control variables in the regression, this counterfactual assumption also extends to the control variables; that is the relationship between the control variables and the enrollment rate would have remained unchanged over time for both the treated and control groups in the absence of the treatment. Essentially, the counterfactual is the relationship would be stable event across control variables.

1.3.7 2.7 Spillover effects

Thus far, we have focused on the impact of PROGRESA on the school enrollment of poor households. Repeat your analysis in 2.5, instead focusing on (a) the impact of PROGRESA on the school enrollment of non-poor households, and (b) the impact of PROGRESA on *other outcomes* of poor households that might plausibly have been affected by the PROGRESA program. * Do you observe any impacts of PROGRESA on the school enrollment of the non-poor? * Regardless of whether you find evidence of spillovers to non-poor, describe one or two reasons why PROGRESA *might* have impacted non-poor households. Give concrete examples based on the context in which PROGRESA was implemented. * Do you observe any impacts of PROGRESA on other aspects of the welfare of poor households?

```
[ ]: # Your code here
# a) The impact of progresas on the school enrollment of non-poor households
diffndiff_tablea = pd.DataFrame(columns = ['1997', '1998'], index = ['Control',
↳ 'Treatment'])
diffndiff_tablea.loc['Control', '1997'] = progresas_df[(progresas_df['poor'] == 0
↳) & (progresas_df['progresas'] == 0) & (progresas_df['year'] == 97)]['sc'].
↳mean()
diffndiff_tablea.loc['Control', '1998'] = progresas_df[(progresas_df['poor'] == 0
↳) & (progresas_df['progresas'] == 0) & (progresas_df['year'] == 98)]['sc'].
↳mean()
diffndiff_tablea.loc['Treatment', '1997'] = progresas_df[(progresas_df['poor'] == 0
↳) & (progresas_df['progresas'] == 1) & (progresas_df['year'] == 97)]['sc'].
↳mean()
diffndiff_tablea.loc['Treatment', '1998'] = progresas_df[(progresas_df['poor'] == 0
↳) & (progresas_df['progresas'] == 1) & (progresas_df['year'] == 98)]['sc'].
↳mean()

#calculate difference and estimator
diff_treated_np = diffndiff_tablea.loc['Treatment', '1998'] - diffndiff_tablea.
↳loc['Treatment', '1997']
diff_ctrl_np = diffndiff_tablea.loc['Control', '1998'] - diffndiff_tablea.
↳loc['Control', '1997']
```

```

diffndiff_estimate_np = diff_treated_np - diff_ctrl_np

# # output results
print('Difference-in-difference estimate of the impact for non-poor households:
↳'+ str(diffndiff_estimate_np))
diffndiff_tablea.head()

```

Difference-in-difference estimate of the impact for non-poor households:
3.4275264633842895e-05

```

[ ]:
      1997      1998
Control  0.762587  0.776337
Treatment 0.795264  0.809049

```

```

[ ]: # b) the impact of PROGRESA on other outcomes of poor households eg) monthly
↳wages of head of household
diffndiff_tableb = pd.DataFrame(columns = ['1997', '1998'], index = ['Control',
↳'Treatment'])
diffndiff_tableb.loc['Control', '1997'] = progres_a_df[(progres_a_df['poor'] ==
↳1) & (progres_a_df['progres_a'] == 0) & (progres_a_df['year'] == 97)]['hohwag'].
↳mean()
diffndiff_tableb.loc['Control', '1998'] = progres_a_df[(progres_a_df['poor'] ==
↳1) & (progres_a_df['progres_a'] == 0) & (progres_a_df['year'] == 98)]['hohwag'].
↳mean()
diffndiff_tableb.loc['Treatment', '1997'] = progres_a_df[(progres_a_df['poor'] ==
↳1) & (progres_a_df['progres_a'] == 1) & (progres_a_df['year'] == 97)]['hohwag'].
↳mean()
diffndiff_tableb.loc['Treatment', '1998'] = progres_a_df[(progres_a_df['poor'] ==
↳1) & (progres_a_df['progres_a'] == 1) & (progres_a_df['year'] == 98)]['hohwag'].
↳mean()

#calculate difference and estimator
diff_treated = diffndiff_tableb.loc['Treatment', '1998'] - diffndiff_tableb.
↳loc['Treatment', '1997']
diff_ctrl = diffndiff_tableb.loc['Control', '1998'] - diffndiff_tableb.
↳loc['Control', '1997']

diffndiff_estimateb = diff_treated - diff_ctrl

# output results
print('Difference-in-difference estimate of the impact on monthly wages: '+
↳str(diffndiff_estimateb))
diffndiff_tableb.head()

```

Difference-in-difference estimate of the impact on monthly wages: 0.0

```
[ ]:
      1997      1998
Control  573.163558  573.163558
Treatment 544.339544  544.339544
```

```
[ ]: # b) the impact of PROGRESA on other outcomes of poor households eg) monthly
      ↪ wages of head of household
diffndiff_tableb = pd.DataFrame(columns = ['1997', '1998'], index = ['Control',
      ↪ 'Treatment'])
diffndiff_tableb.loc['Control', '1997'] = progres_a_df[(progres_a_df['poor'] ==
      ↪ 1) & (progres_a_df['progres_a'] == 0) & (progres_a_df['year'] == 97)]['hohedu'].
      ↪ mean()
diffndiff_tableb.loc['Control', '1998'] = progres_a_df[(progres_a_df['poor'] ==
      ↪ 1) & (progres_a_df['progres_a'] == 0) & (progres_a_df['year'] == 98)]['hohedu'].
      ↪ mean()
diffndiff_tableb.loc['Treatment', '1997'] = progres_a_df[(progres_a_df['poor'] ==
      ↪ 1) & (progres_a_df['progres_a'] == 1) & (progres_a_df['year'] == 97)]['hohedu'].
      ↪ mean()
diffndiff_tableb.loc['Treatment', '1998'] = progres_a_df[(progres_a_df['poor'] ==
      ↪ 1) & (progres_a_df['progres_a'] == 1) & (progres_a_df['year'] == 98)]['hohedu'].
      ↪ mean()

# calculate difference and estimator
diff_treatedb = diffndiff_tableb.loc['Treatment', '1998'] - diffndiff_tableb.
      ↪ loc['Treatment', '1997']
diff_ctrlb = diffndiff_tableb.loc['Control', '1998'] - diffndiff_tableb.
      ↪ loc['Control', '1997']

diffndiff_estimateb = diff_treatedb - diff_ctrlb

# output results
print('Difference-in-difference estimate of the impact on head of household
      ↪ education: ' + str(diffndiff_estimateb))
diffndiff_tableb.head()
```

Difference-in-difference estimate of the impact on head of household education:
0.0

```
[ ]:
      1997      1998
Control  2.590348  2.590348
Treatment 2.663139  2.663139
```

1. Do you observe any impacts of PROGRESA on the school enrollment of the non-poor?
 - Based on the difference-in-difference estimated performed, there seems to be no impact of progres_a on enrollment for non-pool households (dif = 3.4275264633842895e-05)
2. Regardless of whether you find evidence of spillovers to non-poor, describe one or two reasons why PROGRESA *might* have impacted non-poor households. Give concrete examples based on the context in which PROGRESA was implemented.

- Although from our findings on enrollment there was no spillover effect, progresas may have impacted non-poor households whereby the focus on increasing enrollment of poor households could have resulted in changes in educational infrastructure such as improvement of libraries and communal learning spaces. These would serve to benefit non-poor households residing in the same area. Moreover, local economic stimulus from progresas injected money could stimulate economic activity and increased job opportunities and community development projects in areas where non-poor children also reside.
3. Do you observe any impacts of PROGRESA on other aspects of the welfare of poor households?
- There are limited aspects of welfare in this data set and only at 2 time points (1997 and 1998) so when taking a look at the effects on the head of household education level and at the head of household wages, there was absolutely no difference in the estimate of the impact of progresas on these since the data seems to be the same. That's not to say there would have been a change over a longer period of time or by looking at other aspects of welfare.

1.3.8 2.8 Summary

- Based on all the analysis you have undertaken to date, do you believe that Progresas had a causal impact on the enrollment rates of poor households in Mexico?

Discuss your results here - Based on the several analyses we have performed here in this exercise including a difference in difference regression with multiple control variables, we can conclude there is a strong correlation between households that received progresas aid and increased enrollment rate in school. This may have a causal impact although further analyses at later time points following intervention while also considering more related control variables are needed to better assess a causal role. - This correlation makes sense since progresas was a conditional cash transfer program whereby households only received benefits if they engaged in behaviour to improve family welfare, such as children education and health. One of the limitations of this analysis is that it doesn't consider the fine incentives and disincentives for receiving aid, such as termination of program benefits after a certain point in secondary school which would surely impact the enrollment rates if we broke down our analysis by grade. However overall given the information we have here, it appears progresas had an overall positive impact on poor household school enrollment rates.